Scattering by a Central Potential; The Hamiltonian and Hamilton's Equations of Motion

HC 6001 — Dr. Castillo

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This chapter typesets the instructor's handwritten lecture notes for HC 6001 (October 22, 2024) in the notation and equation conventions of Goldstein, Poole & Safko, *Classical Mechanics*, 3rd ed. Two topics are covered: the kinematics and dynamics of scattering by a central potential, culminating in the Rutherford cross section (Sections 1–3), and the passage from the Lagrangian to the Hamiltonian formalism via the Legendre transform, the opening of the course's Chapter 8 (Section 4). Occasional remarks below that refer to "the original notes" point to the instructor's handwritten source pages, reproduced in part as figures throughout.

1 Kinematics of Scattering by a Central Potential

Consider a particle of mass m incident on a fixed central-potential scattering center with energy E, velocity v_0, and angular momentum ℓ:

$$E = \tfrac{m}{2}v_0^2, \qquad v_0 = \sqrt{2E/m}, \qquad \ell = smv_0 = s\sqrt{2mE}, \tag{1}$$

where s is the *impact parameter*. Equivalently, $s = \ell/\sqrt{2mE}$.

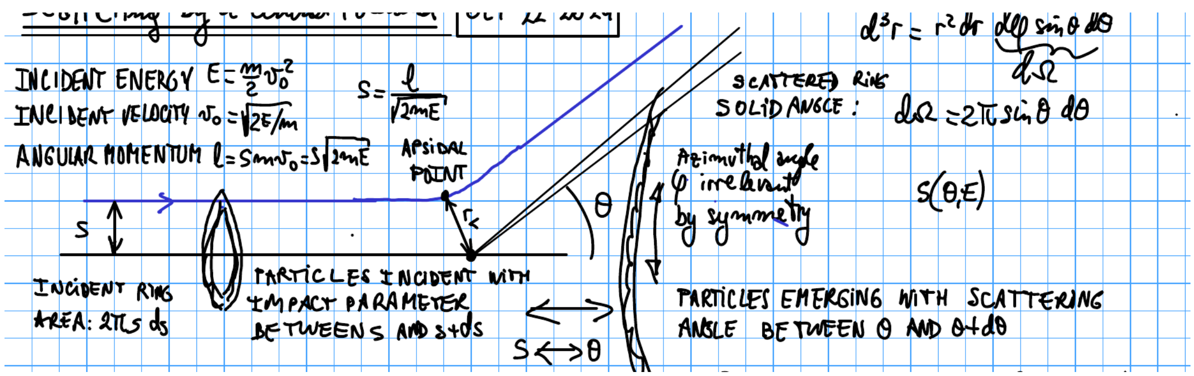

Figure 1: Impact-parameter and scattered-ring geometry: particles incident with impact parameter between s and $s+ds$ (left) emerge with scattering angle between θ and $\theta+d\theta$ into the ring of solid angle $d\Omega$ (right). The azimuthal angle φ is irrelevant by the symmetry of a central potential, so the scattering problem reduces to a function $s = S(\theta, E)$.

As Fig. 1 also records, the solid-angle element used throughout is $d^3r = r^2\,dr\,d\varphi\sin\theta\,d\theta$, with $d\Omega \equiv d\varphi\sin\theta\,d\theta$ integrated over φ to give $d\Omega = 2\pi\sin\theta\,d\theta$ below.

The incident intensity is the number of particles incident per unit time per unit cross-sectional area of the beam,

$$I = \frac{\#\ \text{particles incident per unit time}}{\text{incident beam cross-section area}}.$$

An incident ring of area $2\pi s\, ds$ carries particles with impact parameter between s and $s + ds$; by conservation of particle number these emerge into the solid angle $d\Omega = 2\pi \sin\theta\, d\theta$ subtending the corresponding range of scattering angle:

$$I(2\pi s)\,|ds| \;=\; I\,\sigma(\theta)\,2\pi \sin\theta\,|d\theta|,$$

where $\sigma(\Omega)\,d\Omega$ is, by definition, the number of particles scattered into $d\Omega$ per unit time, divided by I. Hence

$$\boxed{\sigma(\theta) = \frac{s}{\sin\theta}\left|\frac{ds}{d\theta}\right|}, \qquad \boxed{\sigma_T = \int d\Omega\, \sigma(\Omega)}, \tag{2}$$

the *differential* and *total* scattering cross sections.

2 Relation Between Impact Parameter and Scattering Angle

Let ψ be the angle, measured at the apsidal point, between the incoming asymptote and the line to the center of force. Two cases arise, depending on whether the force is repulsive or attractive:

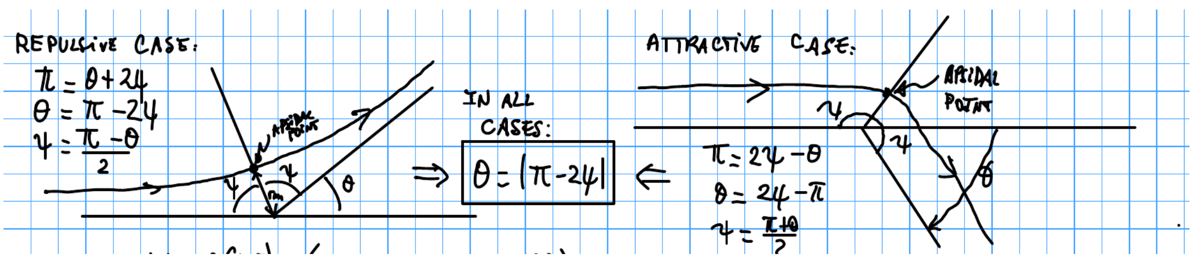

Figure 2: Repulsive case ($\pi = \theta + 2\psi$, left) and attractive case ($\pi = 2\psi - \theta$, right); ψ is the angle at the apsidal point between the incoming asymptote and the line to the force center.

$$\text{Repulsive:} \quad \pi = \theta + 2\psi, \qquad \theta = \pi - 2\psi, \qquad \psi = \frac{\pi - \theta}{2}; \tag{3}$$

$$\text{Attractive:} \quad \pi = 2\psi - \theta, \qquad \theta = 2\psi - \pi, \qquad \psi = \frac{\pi + \theta}{2}. \tag{4}$$

In both cases,

$$\boxed{\theta = |\pi - 2\psi|}. \tag{5}$$

For a general central potential $V(r)$, conservation of energy and angular momentum give

$$E = \frac{m}{2}\dot{r}^2 + V(r) + \frac{\ell^2}{2mr^2}, \qquad \dot{r} = \left[\frac{2}{m}\left(E - V(r) - \frac{\ell^2}{2mr^2}\right)\right]^{1/2},$$

and, with $\ell = mr^2\dot{\theta}$,

$$d\theta = \dot{\theta}\,dt = \frac{\ell}{mr^2}\,\frac{dr}{\dot{r}} = \frac{s\,r^{-2}\,dr}{\left[1 - \dfrac{V(r)}{E} - \dfrac{s^2}{r^2}\right]^{1/2}},$$

using $\ell/\sqrt{2mE} = s$. Integrating from the apsidal distance r_m (where $u_m \equiv 1/r_m$) out to $r \to \infty$, and substituting $u = 1/r$,

$$\psi = \int_0^{u_m} \frac{s\,du}{\left[1 - \dfrac{V(1/u)}{E} - s^2u^2\right]^{1/2}}, \tag{6}$$

so that, combined with Eq. (5),

$$\boxed{\theta(s) = \left|\pi - 2\int_0^{u_m} \frac{s\,du}{\left[1 - \dfrac{V(1/u)}{E} - s^2u^2\right]^{1/2}}\right|}. \tag{7}$$

From $\theta(s)$ one computes $|ds/d\theta| = 1/|d\theta/ds|$ and hence $\sigma(\theta)$ via Eq. (2).

Remark 1. Equations (5)–(7) are valid for *any* central potential $V(r)$; nothing in this section has assumed a specific force law.

3 Coulomb (Rutherford) Scattering

We now specialize the universal result Eq. (7) to the Coulomb interaction between two charges Ze and $Z'e$ (Gaussian units, following Goldstein's convention), with force $f = ZZ'e^2/r^2 = -k/r^2$ and potential $V(r) = -k/r$, $k = -ZZ'e^2$.

For $V(r) = -k/r$ the orbit is a conic section,

$$\frac{1}{r} = C_1\big[1 + \varepsilon\cos(\tilde\theta - \tilde\theta')\big], \qquad C_1 = \frac{mk}{\ell^2}, \qquad \varepsilon = \left[1 + \frac{2E\ell^2}{mk^2}\right]^{1/2} = \left[1 + \left(\frac{2sE}{ZZ'e^2}\right)^2\right]^{1/2} > 1 \tag{8}$$

for $E > 0$, using $\ell = s\sqrt{2mE}$.

Remark 2 (Orbit classification). For this potential the sign of E fixes the orbit type: $E > 0$ gives a hyperbolic orbit ($\varepsilon > 1$), $E = 0$ a parabolic orbit ($\varepsilon = 1$), and $E < 0$ an elliptic orbit ($\varepsilon < 1$). This three-way classification by the sign of E is specific to $V(r) = -k/r$ and does not hold for a general central potential. Scattering states have $E > 0$, so only the hyperbolic case is used below.

Two situations occur:

$$\text{repulsive:}\quad ZZ' > 0 \;\Rightarrow\; k = -ZZ'e^2 < 0, \qquad C_1 < 0; \tag{9}$$

$$\text{attractive:}\quad ZZ' < 0 \;\Rightarrow\; k = -ZZ'e^2 > 0, \qquad C_1 > 0 \quad (\text{also gravity, } k = Gm_1m_2). \tag{10}$$

Repulsive case ($k < 0$). Physical requires $1/r > 0$, i.e. $\varepsilon\cos(\tilde\theta - \tilde\theta') < -1$; since $\tilde\theta = \tilde\theta'$ is unphysical (it would give $1/r = C_1(1+\varepsilon) < 0$), the periapsis is at $\tilde\theta' = \pi$, so $\cos(\tilde\theta - \tilde\theta') = -\cos\tilde\theta$ and

$$\frac{1}{r} = \frac{mZZ'e^2}{\ell^2}\big[\varepsilon\cos\tilde\theta - 1\big],$$

with apsidal angle $\tilde\theta = 0$. The distance of closest approach (minimum r, maximum $1/r$) therefore occurs at $\tilde\theta = 0$, giving $1/r_m = (mZZ'e^2/\ell^2)(\varepsilon - 1)$. Requiring $\psi = \tilde\theta(r \to \infty)$, i.e. $1/r \to 0$, gives $\varepsilon\cos\psi = 1$, and using $\theta = \pi - 2\psi$ (repulsive case) one finds $\dfrac{1}{\varepsilon} = \sin\dfrac{\theta}{2}$.

Attractive case ($k > 0$). By symmetry ($\tilde{\theta}' = 0$ is now the periapsis), the minimum distance is $1/r_m = C_1(1+\varepsilon)$, again at $\tilde{\theta} = 0$. The same asymptotic analysis with $\theta = 2\psi - \pi$ gives $-\dfrac{1}{\varepsilon} = -\sin\dfrac{\theta}{2}$, i.e. the same relation $\dfrac{1}{\varepsilon} = \sin\dfrac{\theta}{2}$.

Combining both cases. Since $\cot^2(\theta/2) = 1/\sin^2(\theta/2) - 1 = \varepsilon^2 - 1$, and from Eq. (8), $\varepsilon^2 - 1 = (2sE/ZZ'e^2)^2$, one obtains the impact parameter as a function of scattering angle,

$$\boxed{s = S(\theta, E) = \left|\frac{ZZ'e^2}{2E}\right| \cot\frac{\theta}{2}}. \tag{11}$$

Differentiating,

$$\frac{ds}{d\theta} = -\frac{1}{2}\left|\frac{ZZ'e^2}{2E}\right| \csc^2\frac{\theta}{2},$$

and substituting s and $|ds/d\theta|$ into Eq. (2), using $\sin\theta = 2\sin(\theta/2)\cos(\theta/2)$,

$$\sigma(\theta) = \frac{1}{\sin\theta}\left|\frac{ZZ'e^2}{2E}\right|^2 \cot\frac{\theta}{2}\cdot\frac{1}{2}\csc^2\frac{\theta}{2} = \left(\frac{ZZ'e^2}{2E}\right)^2 \frac{1}{4}\csc^4\frac{\theta}{2},$$

i.e. the **Rutherford scattering cross section** for Coulomb forces:

$$\boxed{\sigma(\theta) = \frac{1}{4}\left(\frac{ZZ'e^2}{2E}\right)^2 \csc^4\frac{\theta}{2}}. \tag{12}$$

Remark 3. The cross section (12) is the same for the repulsive and attractive cases, since it depends on ZZ' only through $(ZZ')^2$.

Remark 4. It diverges like E^{-2} at low energies.

Remark 5. It diverges like θ^{-4} (*very strong*) at small scattering angles.

Remark 6. This classical result is exactly the same as the result obtained in quantum mechanics.[1]

Remark 7 (Notation). Following Goldstein's convention, the differential and total cross sections are written $\sigma(\Omega)$ and σ_T; other references commonly write $d\sigma/d\Omega$ and σ for the same two quantities, respectively.

3.1 Total Scattering Cross Section

With $\sigma(\theta) = \frac{1}{4}(ZZ'e^2/2E)^2\csc^4(\theta/2)$ and $d\Omega = 2\pi\sin\theta\,d\theta = 4\pi\sin(\theta/2)\cos(\theta/2)\,d\theta$,

$$\sigma_T = \frac{1}{4}\left(\frac{ZZ'e^2}{2E}\right)^2\cdot 4\pi\int_0^\pi \frac{\sin(\theta/2)\cos(\theta/2)}{\sin^4(\theta/2)}\,d\theta = \pi\left(\frac{ZZ'e^2}{2E}\right)^2\int_0^\pi \frac{\cos(\theta/2)}{\sin^3(\theta/2)}\,d\theta \sim \int_0 \frac{d\theta}{(\theta/2)^3},$$

which is **strongly divergent** as $\theta \to 0$: classically, $\sigma_T = +\infty$ for any long-range potential; quantum mechanically, $\sigma_T = +\infty$ as well for potentials $V(r)$ that do not decay fast enough as $r \to \infty$.

Caveat 1. In a dense, conducting material the bare Coulomb potential $V = -k/r$ is replaced by the screened ("Yukawa") potential

$$V^{\text{screen}}(r) = -\frac{k}{r}\,e^{-r/\lambda},$$

which regularizes the divergence. In non-conducting materials $V^{\text{screen}}(r)$ may take a different functional form, but will still vanish much faster than $1/r$ as $r \to \infty$.

[1]Editorial note, not in the original notes: the first Born approximation for the Coulomb potential reproduces Eq. (12) exactly, which is the standard explanation for why the classical and quantum cross sections coincide here.

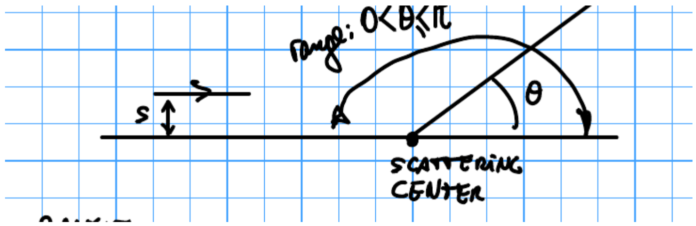

Figure 3: Scattering-center geometry: a particle with impact parameter between s and $s + ds$, incident along the horizontal line, is deflected by angle θ at the scattering center.

4 The Hamiltonian and Hamilton's Equations of Motion

We now turn to a different topic, which opens the course's Chapter 8 immediately following the scattering material of Sections 1–3 in the same lecture: rather than continuing with the Lagrangian description $L(q, \dot{q}, t)$ used implicitly above, we develop an equivalent formulation in terms of the generalized coordinates and their conjugate momenta, (q, p, t). Starting from the Lagrangian $L = T - V = L(q_1, \ldots, q_n, \dot{q}_1, \ldots, \dot{q}_n, t)$ and the Euler–Lagrange equations

$$0 = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i}, \qquad p_i \equiv \left.\frac{\partial L}{\partial \dot{q}_i}\right|_{q_j, \dot{q}_j},$$

Remark 8 (Midterm remark — an invalid claim). In plane polar coordinates it is tempting to write $\partial L/\partial\dot{\theta} = p_\theta = \ell_z$ and conclude $L = L(r, \dot{r}, p_\theta)$, i.e. to substitute the canonical momentum p_θ directly back into the Lagrangian in place of $\dot{\theta}$. This step is **invalid**: the Lagrangian is a function of the generalized coordinates and *velocities*, $L = L(q, \dot{q}, t)$; replacing $\dot{q}$ by its conjugate momentum p changes the independent variables and is precisely the Legendre transformation carried out below to pass from L to H — it cannot be done term-by-term inside L itself. Consequently the crossed-out "⇒ Lagrange Eqn." that follows in the original notes is equally invalid: substituting the ill-defined $L(r, \dot{r}, p_\theta)$ into the Euler–Lagrange equation would silently discard the r-dependence hidden inside $p_\theta = mr^2\dot{\theta}$. (Marked as crossed out in the original notes as a caution against this common exam error.)

4.1 The Legendre Transformation

The passage from $L(q, \dot{q}, t)$ to a new function of (q, p, t) is an instance of the general Legendre transformation. For $f(x, y)$ with $df = \dfrac{\partial f}{\partial x}dx + \dfrac{\partial f}{\partial y}dy \equiv u\,dx + v\,dy$, define $g = f - ux$; then

$$dg = df - d(ux) = u\,dx + v\,dy - u\,dx - x\,du = -x\,du + v\,dy,$$

so $g = g(u, y)$: the transformation trades the variable x for the conjugate variable $u = \partial f/\partial x$.

Remark 9 (Thermodynamic analogy). The same transformation appears in thermodynamics. With the first law $dU = \delta Q + \delta W = T\,dS - p\,dV$ (fluid at fixed particle number N), $U = U(S, V)$ is the internal energy. To use (T, V) as independent variables instead, define the Helmholtz free energy $A = U - TS$:

$$dA = dU - d(TS) = (T\,dS - p\,dV) - T\,dS - S\,dT = -S\,dT - p\,dV \;\Rightarrow\; A = A(T, V),$$

exactly the Legendre transform $g = f - ux$ with $(x, u) \to (S, T)$.

4.2 From the Lagrangian to the Hamiltonian

Apply the Legendre transform to $L(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n, t)$, trading each $\dot{q}_i$ for its conjugate momentum $p_i = \partial L/\partial \dot{q}_i$. Define

$$H \equiv -\Big(L - \sum_i p_i \dot{q}_i\Big) = \sum_i p_i \dot{q}_i - L.$$

Then

$$dH = -dL + \sum_i d(p_i \dot{q}_i) = -\sum_i \left(\frac{\partial L}{\partial q_i} dq_i + \frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i\right) - \frac{\partial L}{\partial t} dt + \sum_i \big(p_i \, d\dot{q}_i + \dot{q}_i \, dp_i\big),$$

and since $\partial L/\partial \dot{q}_i = p_i$ the $d\dot{q}_i$ terms cancel:

$$dH = -\frac{\partial L}{\partial t} dt + \sum_i \left(\dot{q}_i \, dp_i - \frac{\partial L}{\partial q_i} dq_i\right) \implies H = H(p_1, \dots, p_n, q_1, \dots, q_n, t).$$

Using the Euler–Lagrange equation, $\partial L/\partial q_i = \dot{p}_i$, and comparing the two expressions for dH term by term gives **Hamilton's canonical equations of motion**:

$$\boxed{\frac{\partial H}{\partial p_i} = \dot{q}_i, \qquad \frac{\partial H}{\partial q_i} = -\frac{\partial L}{\partial q_i} = -\dot{p}_i, \qquad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}.} \tag{13}$$